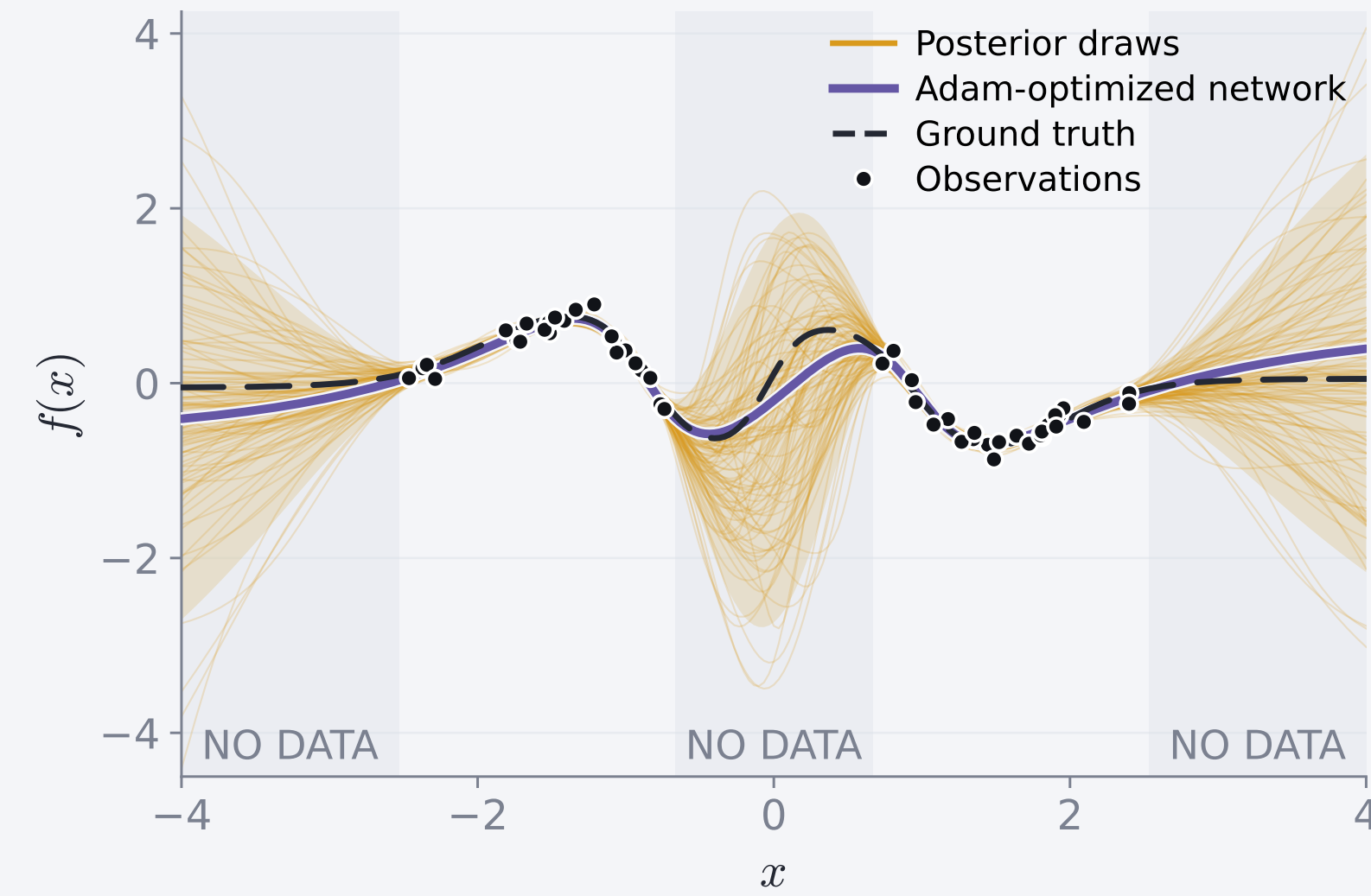




Why sample a neural network?

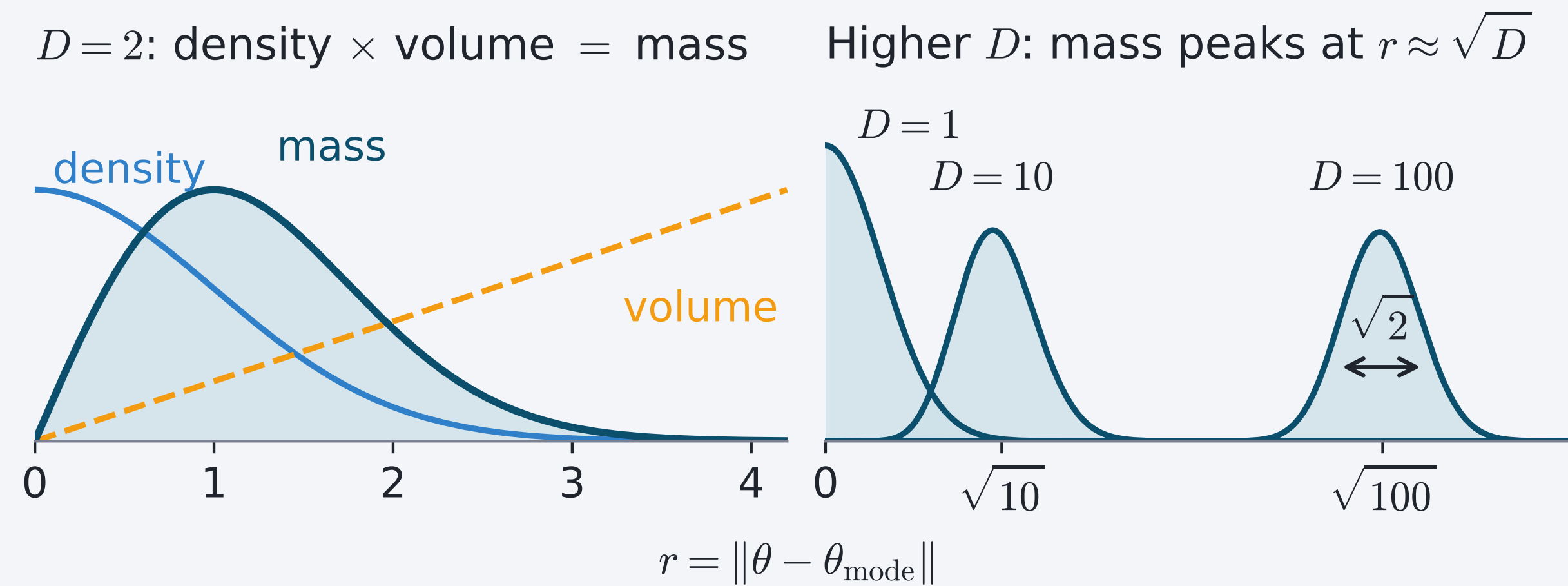


Ordinary training: one set of weights $\rightarrow$ one function.

Bayesian sampling: a distribution over weights $\theta \rightarrow$ uncertainty over functions.

$$\pi(\theta) = p(\theta | \mathcal{D}) \propto \underbrace{p(\mathcal{D} | \theta)}_{\text{fit}} \underbrace{p(\theta)}_{\text{prior}}$$

Where the probability lives



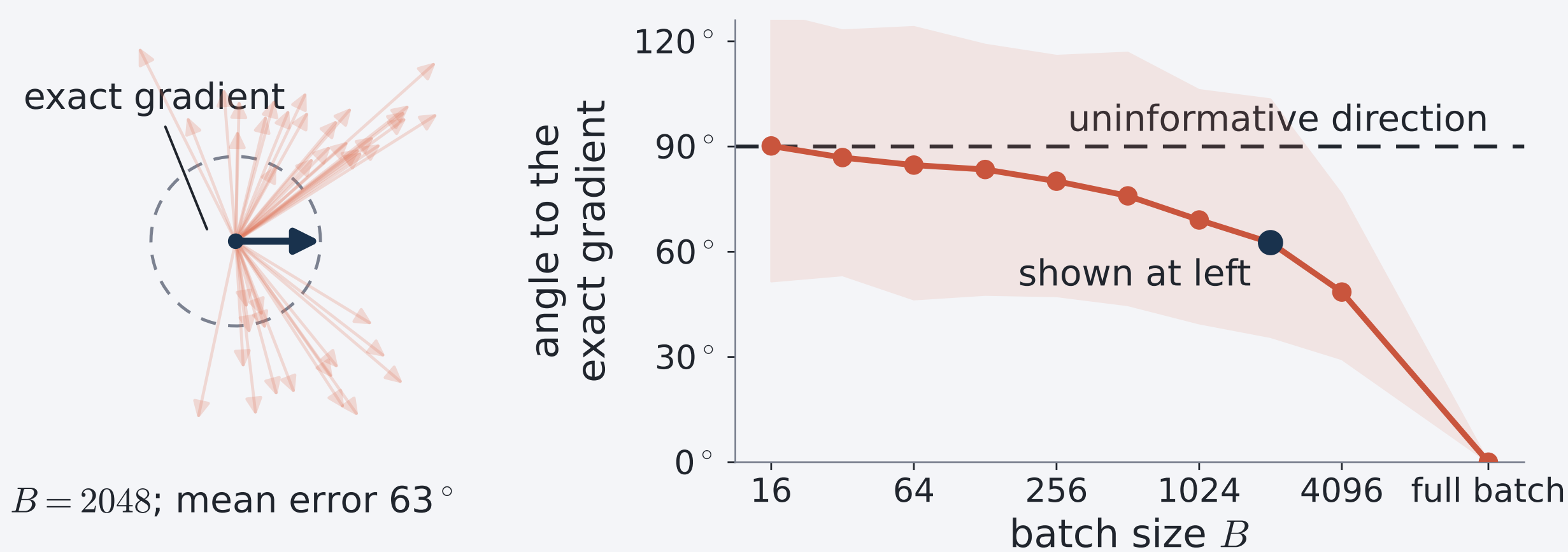
Because volume grows as r^{D-1} , probability concentrates in the *typical set*: a thin shell near $\|\theta\|^2 \approx D$. A correct sampler therefore spends almost all its time on this shell.

MCMC in the era of big data

Markov chain Monte Carlo (MCMC) generates a sequence of correlated samples targeting π . Gradient-based samplers use $\nabla \log \pi(\theta)$ to guide their updates. Computing this gradient exactly uses all N data points. A minibatch of size $B \ll N$ is unbiased and roughly N/B times cheaper, but never exact:

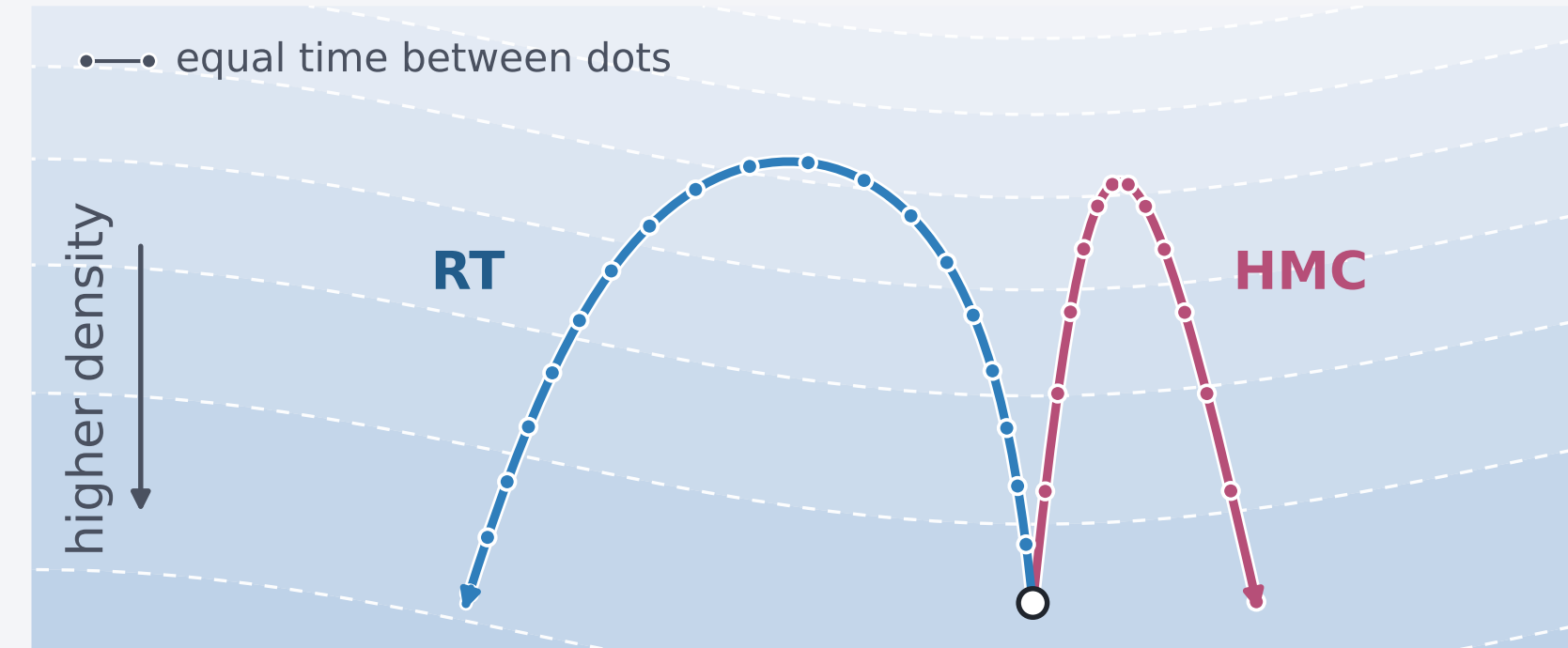
$$\hat{g}_B(\theta) = \nabla \log p(\theta) + \frac{N}{B} \sum_{i \in B} \nabla \log p(\mathcal{D}_i | \theta), \quad \text{Cov}[\hat{g}_B] \propto 1/B.$$

Example: UCI Superconductivity Bayesian linear regression ($D = 81$, $N = 17,010$).



HMC and ray tracing under gradient noise

Hamiltonian Monte Carlo (HMC)^[1] and ray tracing (RT)^[2] both use the log-posterior gradient to build trajectories. HMC treats it as a force, changing a particle's speed and direction. RT treats the posterior as a refractive medium; its ray moves at unit speed, so the gradient changes only direction.

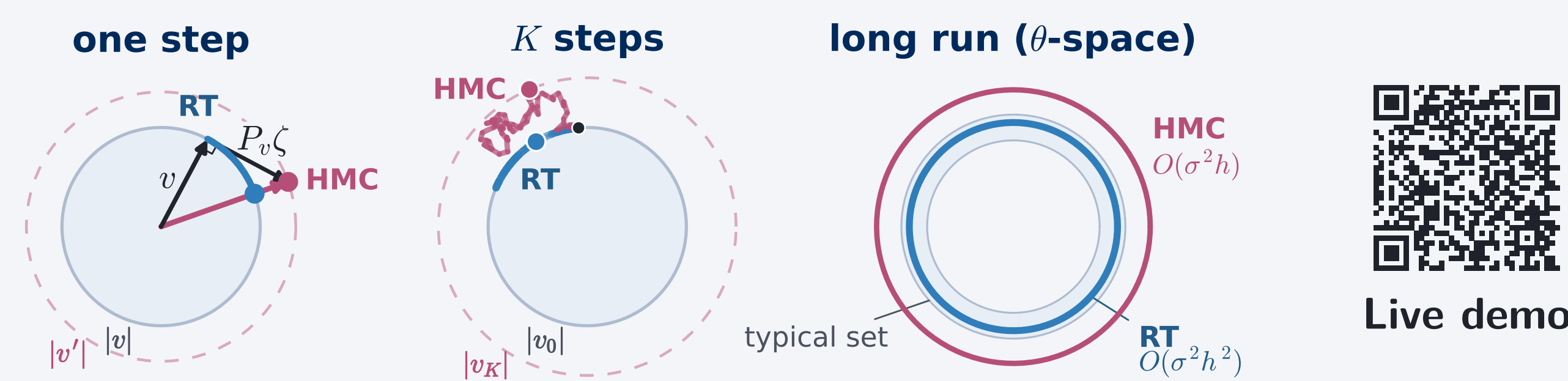


1. One noisy step

Without a Metropolis correction, mean-zero noise raises HMC's expected kinetic energy and inflates its speed. For RT, with isotropic error $\sigma \zeta$ and $n(\theta) = \pi(\theta)^{1/(D-1)}$, the unit direction obeys

$$\frac{du}{ds} = P_u \left(\nabla \log n(\theta) + \frac{\sigma}{D-1} \zeta \right), \quad P_u = I - uu^\top.$$

P_u projects onto the tangent plane, so noise changes u 's direction but not its norm.



2. Along a trajectory

With step size h and trajectory length S , the S/h noisy kicks accumulate as $\frac{S}{h} \cdot O(\sigma^2 h^2) = O(\sigma^2 h)$. Let $\sigma_c(h) \propto h^{-p}$ be the largest noise scale keeping long-run bias below a fixed tolerance. Unadjusted minibatch HMC has $p = \frac{1}{2}$.

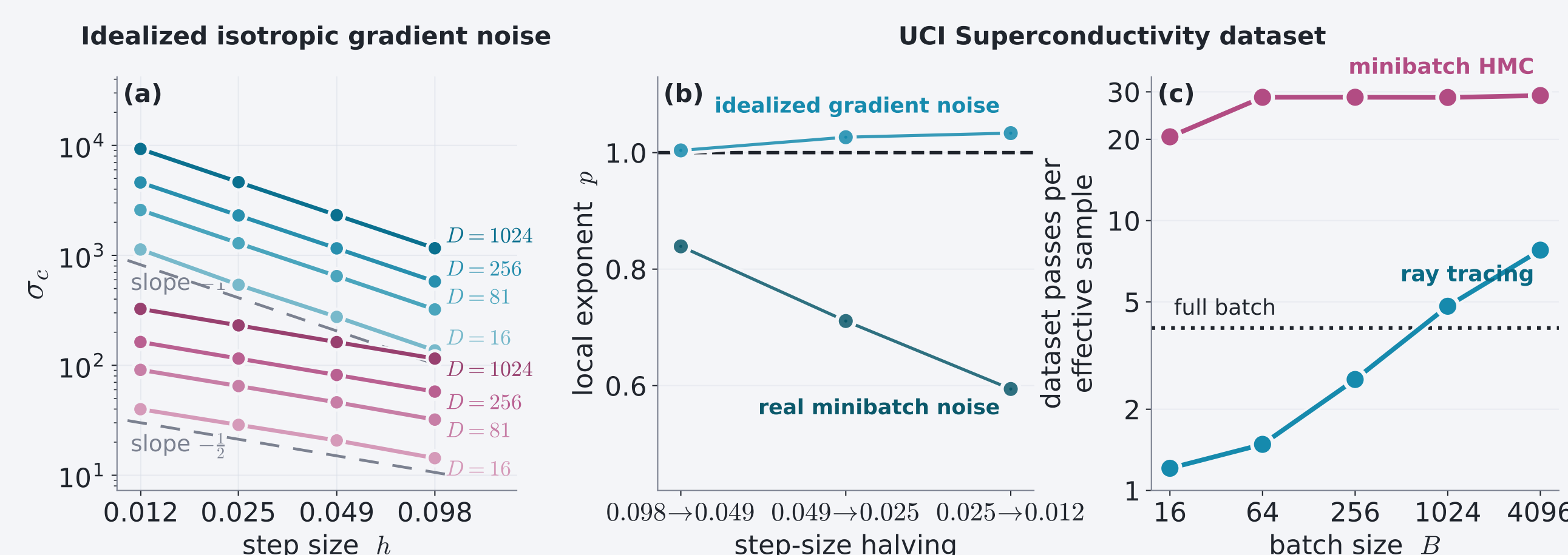
3. Long-run bias

For RT, isotropic rotations leave the uniform direction distribution unchanged, so the $O(\sigma^2 h)$ effect creates no leading bias:

$$b_{\text{RT}}(h, \sigma) = O(\sigma^2 h^2) + o(\sigma^2) \Rightarrow \sigma_c(h) = \Omega(h^{-1}), \quad p \geq 1$$

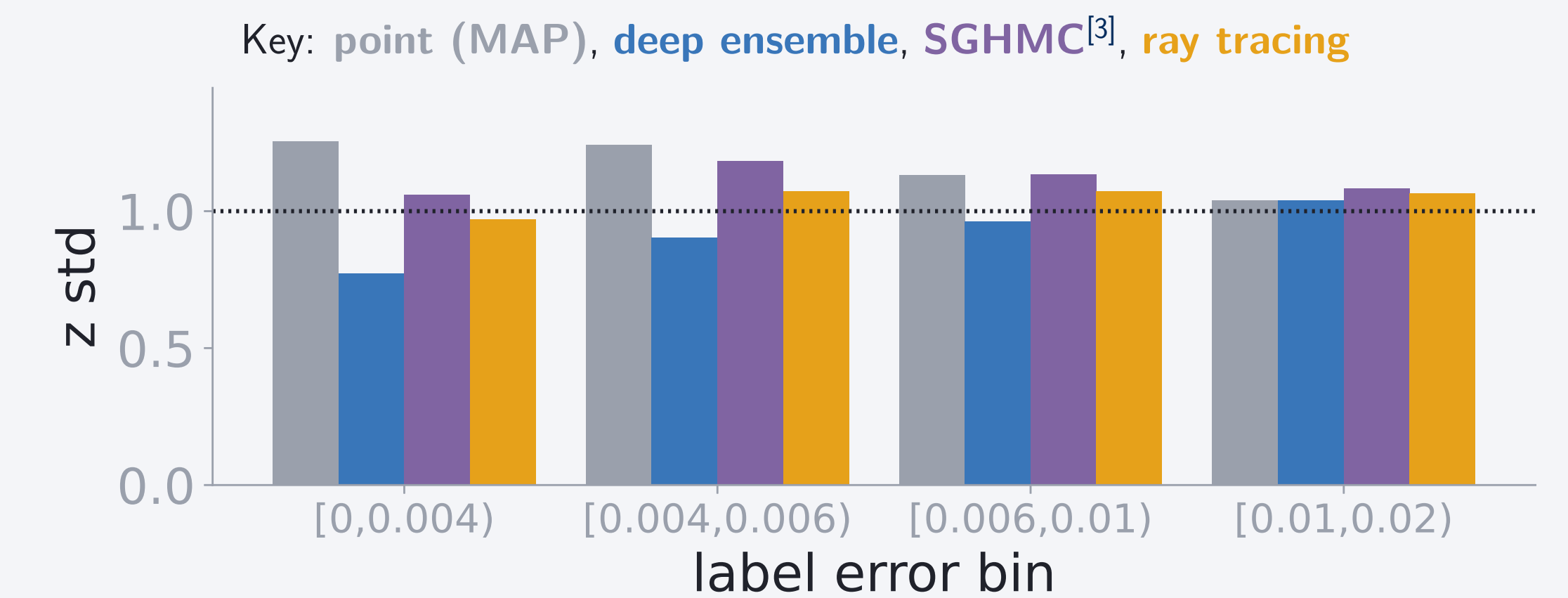
Measured noise tolerance and sampling cost

Across $D = 16$ – 1024 : $\sigma_c \propto h^{-1}$ (RT) versus $h^{-1/2}$ (HMC); real minibatch noise weakens RT's scaling.

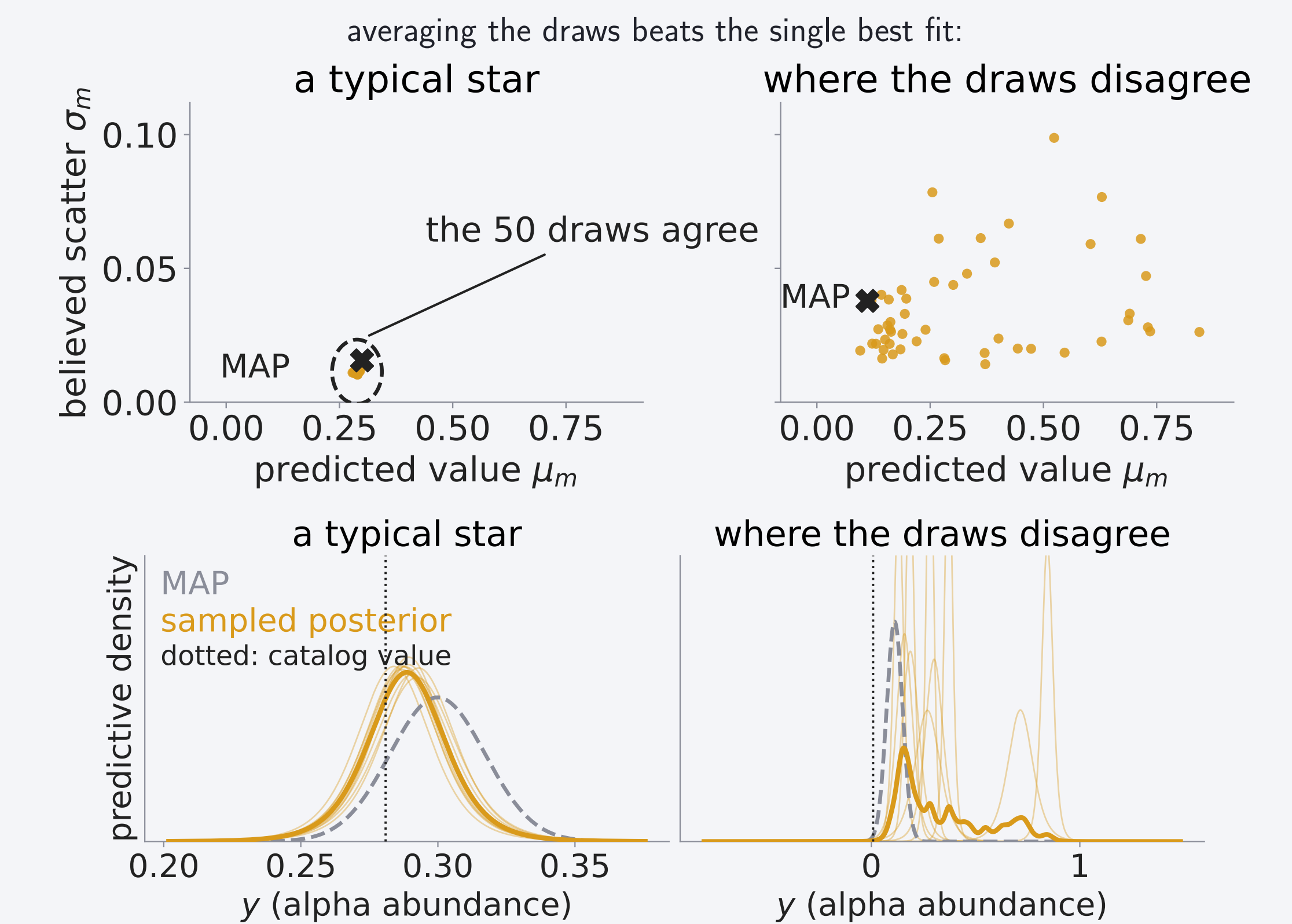


Empirical study on real Gaia data

We predict each star's alpha abundance plus a per-star scatter from its Gaia XP spectrum (110 coefficients), for 126,156 giants with APOGEE labels^[4,5], using a $D = 11,394$ network sampled with minibatch ray tracing.



z std = actual / claimed error: closer to 1 is better.



Conclusion

- Ray tracing moves at fixed speed, so gradient noise changes the ray's direction, not its speed. Under an idealized isotropic model of this noise, halving the step size doubles the tolerated noise level. Since minibatch noise scales as $B^{-1/2}$, this would allow the batch to be four times smaller at the same bias tolerance.
- Real minibatch noise departs from this model, weakening the ideal h^{-1} scaling. Even so, smaller steps tolerate more noise, and smaller batches reduce RT's cost per effective sample.
- On real stellar data, ray tracing provides well-calibrated uncertainty estimates and competitive predictive performance. Averaging 50 draws gives 27% higher predictive probability on held-out stars than the best single fit.

References

- [1] Betancourt, M. 2017, *A Conceptual Introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434
- [2] Behroozi, P. 2026, *The Ray Tracing Sampler: Bayesian Sampling of Neural Networks for Everyone*, Open Journal of Astrophysics 9, doi:10.33232/001c.165932
- [3] Chen, T., Fox, E. B., Guestrin, C. 2014, *Stochastic Gradient Hamiltonian Monte Carlo*, ICML, PMLR 32(2), 1683
- [4] Gaia Collaboration, Vallenari, A., et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1
- [5] Laroche, A., Speagle, J. S. 2025, *Closing the Stellar Labels Gap: Stellar Label Independent Evidence for $[\alpha/M]$ Information in Gaia BP/RP Spectra*, ApJ 979, 5